605.617 Introduction to GPU Programming

CUDA Streams and Events

Create a program that demonstrates understanding of CUDA Streams and Events. It should perform non-trivial operations, such as performing a variety of mathematical calculations or something even more elaborate. There should be a demonstrable use of streams and events and if timing is measured and presented, there will be more credit given to that.

For your assignment submission, you will need to include a link to the code repository (main branch), which will include an assignment_config.yaml configuration file in the root of the project, and build.sh and run.sh files for building and running your code in a child directory, referenced in the configuration file.

If you can show how this will be utilized in your final project, you will get a one-time bonus of 5%.

Task	% of grade
Create a program that shows usage of CUDA Streams and Events.	50
Test harness executes two separate kernels (or two executions of the same kernel) using CUDA streams and events	10
Output timing or other metrics for comparison of different datasets	10
Quality of your code, measured by use of constants, well-named variables and functions, and useful comments in code	20
Assignment configuration, build.sh, and run.sh	10
One-time bonus: Your submission will be part of your final project	5